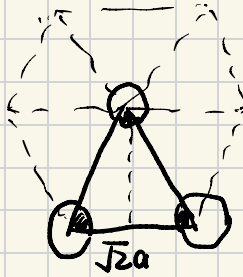
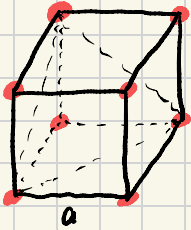


HW 1 何衍宇 CHE Yanning) 12311512

1. Calculate surface density of atom on shaded plane  
a) simple cubic



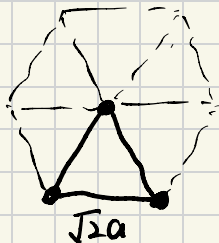
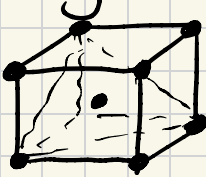
$$h = \frac{\sqrt{6}}{2} a$$

$$\Rightarrow L = \sqrt{2} a$$

$$\text{NO. of atom} = 3 \times \frac{1}{6}$$

$$\begin{aligned} \text{Surface density} &= \frac{\text{NO. of atoms}}{\frac{1}{2} \cdot L \cdot h} \\ &= \frac{3 \times \frac{1}{6}}{\frac{1}{2} \sqrt{2} \frac{\sqrt{6}}{2} a^2} \\ &= \frac{\frac{1}{2}}{\frac{1}{2} \sqrt{3} a^2} \\ &= \frac{1}{\sqrt{3} \cdot a^2} \end{aligned}$$

b) body-centered cubic



$$L = \sqrt{2} a$$

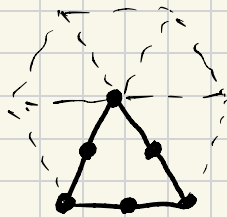
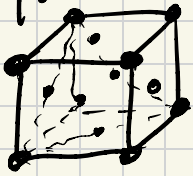
$$\Rightarrow h = \frac{\sqrt{6}}{2} a$$

$$\text{NO. of atom} = 3 \times \frac{1}{6}$$

Hint: body-center atom  
is not on the (1,1,1)

$$\begin{aligned}
 \text{Surface Density} &= \frac{\text{NO. of atom}}{\frac{1}{2} \cdot l \cdot h} \\
 &= \frac{3 \times \frac{1}{6}}{\frac{1}{2} \cdot \sqrt{2} \cdot \frac{\sqrt{6}}{2} a^2} \\
 &= \frac{\frac{1}{2}}{\frac{1}{2} \cdot \sqrt{3} a^2} \\
 &= \frac{1}{\sqrt{3} a^2}
 \end{aligned}$$

c) face-centered cubic



$$\begin{aligned}
 l &= \sqrt{2} a \\
 h &= \frac{\sqrt{6}}{2} a
 \end{aligned}$$

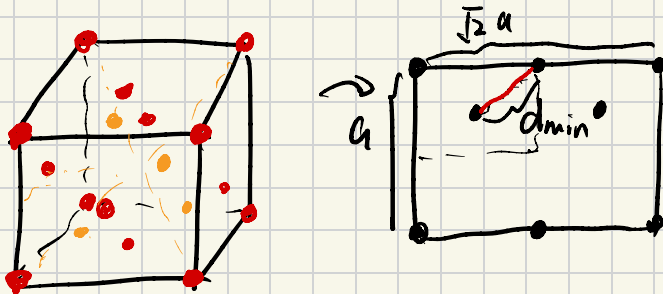
$$\text{NO. of atom} = 3 \times \frac{1}{2} + 3 \times \frac{1}{6}$$

$$\begin{aligned}
 \text{Surface density} &= \frac{\text{NO. of atom}}{\frac{1}{2} \cdot l \cdot h} \\
 &= \frac{3 \times \frac{1}{2} + 3 \times \frac{1}{6}}{\frac{1}{2} \cdot \sqrt{2} \cdot \frac{\sqrt{6}}{2} a^2} \\
 &= \frac{4}{\sqrt{3} a^2}
 \end{aligned}$$

2. silicon crystal  $\rightarrow$  lattice constant:  $5.43 \text{ \AA}$

a) Number =  $8 \times \frac{1}{8} + 4 + 6 \times \frac{1}{2} = 8$

b)



$$d_{\min} = \frac{1}{2} \cdot \sqrt{\left(\frac{\sqrt{2}a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \frac{\sqrt{3}}{4}a$$

$$\Rightarrow d_{\min} \approx 2.35 \text{ \AA}$$

c) Volume density =  $\frac{\text{NO. of atom}}{a^3}$

$$= \frac{8}{(5.43 \times 10^{-8} \text{ cm})^3}$$

$$= 5.0 \times 10^{22} \text{ atoms/cm}^3$$

d) mass density =  $\frac{5.0 \times 10^{22}}{6.02 \times 10^{23}} \times 28.09$

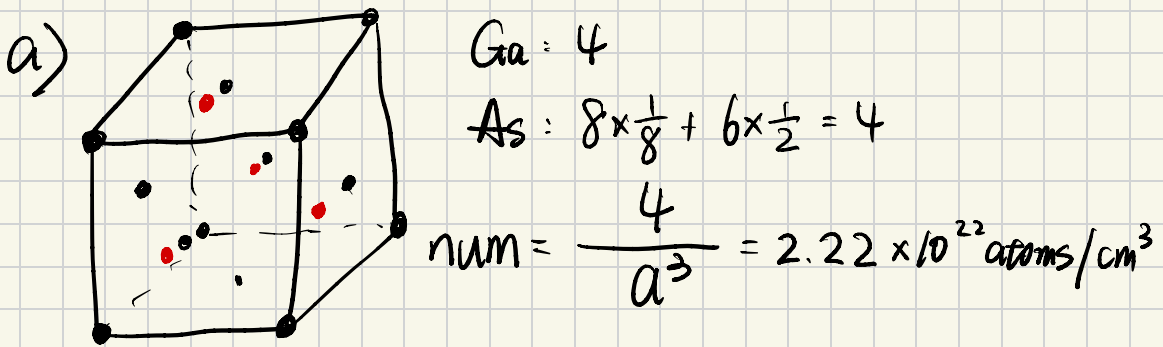
$$= 2.33 \text{ g/cm}^3$$

e) Valence electron density

Valence electron of one Si atom : 4

$$\Rightarrow \text{Valence electron density} = 4 \times 5.0 \times 10^{22} \\ = 2.0 \times 10^{23} \text{ electrons/cm}^3$$

3. Lattice constant  $a = 5.65 \text{ \AA}$  GaAs



$$\text{Ga} : 2.22 \times 10^{22} \text{ atoms/cm}^3$$

$$\text{As} : 2.22 \times 10^{22} \text{ atoms/cm}^3$$

$$\text{b) mass density} = \frac{2.22 \times 10^{22}}{6.02 \times 10^{23}} \times (69.7 + 74.9) \\ = 5.33 \text{ g/cm}^3$$

c) valence electron  $\Rightarrow$  Ga : 3, As : 5

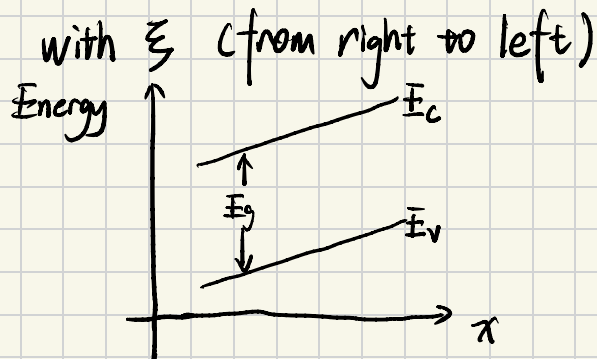
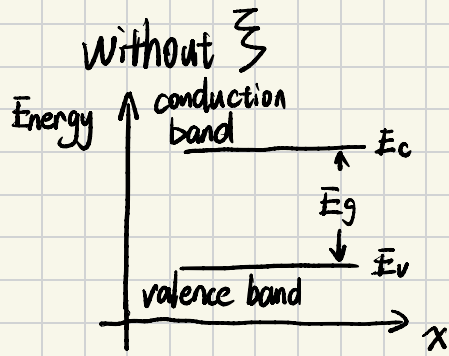
$$\text{valence electron density} = 2.22 \times 10^{22} \times (3 + 5) \\ = 1.78 \times 10^{23} \text{ electrons/cm}^3$$



4. GaAs energy bandgap :  $1.42 \text{ eV} \Rightarrow \text{max}$

$$\begin{cases} E = h\nu \\ c = \lambda\nu \end{cases} \Rightarrow \lambda = \frac{h \cdot c}{E} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{1.42 \times 1.6 \times 10^{-19}} \Rightarrow \lambda \approx 875 \text{ nm}$$

5. Draw:

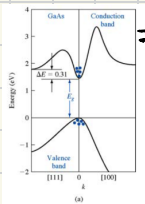


Show:  $F = q \cdot \xi$  ,  $F = \frac{dE}{dx}$  consider as scalar

$$\Rightarrow \xi = \frac{F}{q} = \frac{1}{q} \cdot \frac{dE}{dx}$$

6. Explain

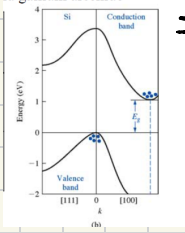
Direct bandgap semiconductor (GaAs)



$\Rightarrow$  The conduction band minimum and the valence band maximum occur at same wavevector  $k$

- Electron recombine with hole by emitting photon
- No momentum change is needed, transition is efficient

# Indirect bandgap semiconductor (Si)



⇒ The conduction band minimum and the valence band maximum are located at different values of  $k$

Recombination: both energy and momentum conservation must be satisfied.  
phonon must assist in process, since photon carry no momentum  
radiative recombination less probable

Reason :

For indirect bandgap semiconductor, the radiative recombination is less efficient and it usually involves phonons. So non-radiative processes dominate and the energy is mostly released as heat rather than light.

So the indirect bandgap semiconductor is not suitable for light emitting devices, such as LED, laser.